

QCD 2025

problem set 5

1. Consider an integral that naively is equal to zero

$$\int_{-\infty}^{\infty} dx [f(x+a) - f(x)] \quad (1)$$

where f is a function that does not vanish at infinity:

$$f(\pm\infty) \neq 0. \quad (2)$$

Calculate (1) expanding in a up to a^2 . What happens when $f'(\pm\infty) = 0$. Generalize this result to the n -dimensional Euclidean integral

$$\Delta(\vec{a}) = \int d^n \vec{r} [f(\vec{r} + \vec{a}) - f(\vec{r})].$$

2. Let's define γ_5 matrix as

$$\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3.$$

Prove that γ_5 is hermitian. Compute anticommutator $\{\gamma_5, \gamma^\mu\}$.

3. Using Dirac equation

$$(\not{p} - m)u(p) = 0 \quad \text{or} \quad \bar{u}(p)(\not{p} - m) = 0$$

prove that

$$q_\mu \bar{u}(p')\gamma^\mu u(p) = 0, \quad q_\mu \bar{u}(p')\gamma^\mu \gamma_5 u(p) = 2m \bar{u}(p')\gamma_5 u(p),$$

where $q = p' - p$. Last equation suggests that for massless fermions there exists a conserved current

$$j_5^\mu = \bar{u}(p')\gamma^\mu \gamma_5 u(p)$$

called axial (or axial-vector) current.

4. To avoid certain confusions concerning γ_5 matrix convince yourself by a direct computation using gamma matrices with upper indices in the Bjorken representation that

$$\text{Tr}[\gamma_5 \gamma^0 \gamma^1 \gamma^2 \gamma^3] = -4i.$$

This suggests that

$$\text{Tr}[\gamma_5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] = -4i\epsilon^{\mu\nu\rho\sigma}$$

where

$$\epsilon^{0123} = 1.$$

When we lower the indices, there are always three space like indices that change sign, and one time-like index that does not. Therefore

$$\begin{aligned}\text{Tr}[\gamma_5 \gamma_\mu \gamma_\nu \gamma_\rho \gamma_\sigma] &= -\text{Tr}[\gamma_5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] \\ &= 4i\epsilon^{\mu\nu\rho\sigma} \\ &= -4i\epsilon_{\mu\nu\rho\sigma}\end{aligned}$$

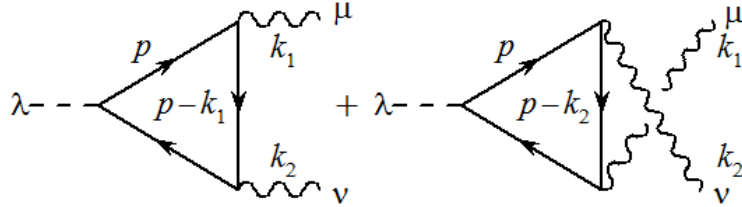
where

$$\epsilon_{0123} = -1.$$

Calculate the following traces:

$$\text{Tr} [\not{p} \gamma_\lambda \gamma_5 \not{p} \gamma_\nu \not{p} \gamma_\mu], \text{Tr} [\not{p} \gamma_5 \gamma_\nu (\not{p} - \not{k}) \gamma_\mu].$$

5. Consider the following loop contribution to the decay of an axial-vector current $\gamma_\lambda \gamma_5$, denoted by a dashed line, to two photons:



Because photons are indistinguishable bosons, the loop correction is a sum of two diagrams.

Using Feynman rules write the explicit expression for the above proces. Note that fermion loop corresponds to the trace over Dirac matrices, and that gamma structures under the trace $\text{Tr}(\dots)$ are written from left to right in the direction *opposite* to the fermion arrows. Use $q = k_1 + k_2$.